

Li Jiaru

(+65) 8039 4376 | jiaru.li@u.nus.edu | github.com/Opsimathy

EDUCATION

National University of Singapore

Master of Computing, General Track

Aug. 2025 – Jan. 2027 (Expected)

Singapore

Imperial College London

Bachelor of Science in Mathematics (Upper Second Class Honours)

Oct. 2022 – Jun. 2025

London, UK

- Course Highlights: Principles of Programming, Network Science, Numerical Analysis, Optimisation, Scientific Computation, Methods for Data Science, Introduction to Game Theory, Classical Dynamics, Quantum Mechanics, etc.

RESEARCH EXPERIENCE

Group Research Project

Imperial College London

Jun. 2024

London, UK

- Title: Scientific Machine Learning: Neural Networks and Neural Differential Equations [[Link](#)].
- Explored both theoretical foundations and practical applications of selected scientific machine learning models, integrating continuous-time dynamical systems with modern neural network architectures.
- Investigated neural networks, neural ODEs and neural CDEs, analysing the history, formulation, training dynamics, advantages and disadvantages especially for modelling irregularly sampled time-series data.
- Served as group leader, coordinating directions, delegating tasks, and facilitating communication within the team.
- Co-authored and co-edited the report and the presentation, contributing to both the writing and the formatting.

Individual Research Project

Imperial College London

Jun. 2023

London, UK

- Title: Diffie–Hellman Key Exchange: From 2 Parties to n Parties [[Link](#)].
- Conducted an independent research project in cryptography on a generalisation of the Diffie–Hellman key exchange protocol, utilising tools from group theory and modular arithmetic.
- Proposed a divide-and-conquer improvement, reducing the computational complexity from $O(n^2)$ to $O(n \log n)$.
- Produced and presented a research poster summarising theoretical results and algorithmic design.

WORK EXPERIENCE

Full-Stack Developer Internship

Sonli Holding Group Co., Ltd.

Jul. 2024 – Sep. 2024

Qingdao, China

- Developed and maintained digital solutions for smart parking and smart community applications.
- Built features across frontend and backend systems, contributing to improved operational efficiency and user experience.
- Collaborated with cross-functional teams to deploy and monitor production systems.

Finance Internship

China International Capital Corporation (CICC)

Jul. 2023 – Sep. 2023

Qingdao, China

- Familiarised with basic financial research, data collection, and presentation preparation.
- Built foundational skills in Microsoft Office, finance, and professional communication.
- Gained hands-on exposure to team workflow in a fast-paced finance environment.

AWARDS & HONOURS

UK University Integration Bee

Feb. 2023

- Obtained third place in the final round of the contest across a number of teams from UK universities, leading the university's team as a first year student [[Link](#)].

National Olympiad in Informatics in Provinces (NOIP)

Nov. 2017

- Obtained first prize as a grade 8 student in the junior group of the competitive programming (OI) competition for high school students at provincial level in China [[Link](#)].

SKILLS

Programming: Python, C/C++, R, Julia, SQL (Postgres), $\text{\LaTeX}$

Libraries: numpy, pandas, matplotlib, seaborn, scikit-learn, TensorFlow, PyTorch, NetworkX

Languages: Chinese (native), English (proficient), Japanese (intermediate)